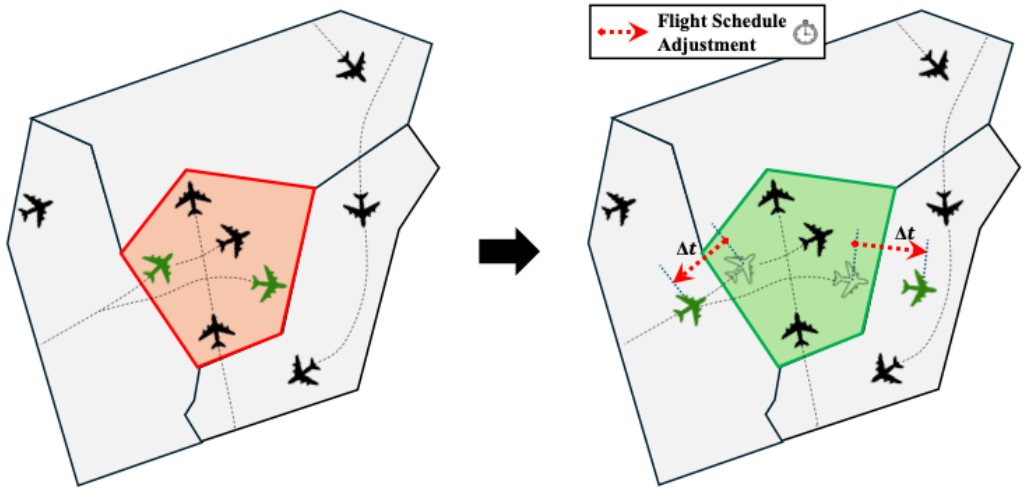
Game-theoretic Decentralized Coordination for Airspace Sector Overload Mitigation

Sector Overload Problem in Noncooperative Air Traffic Management

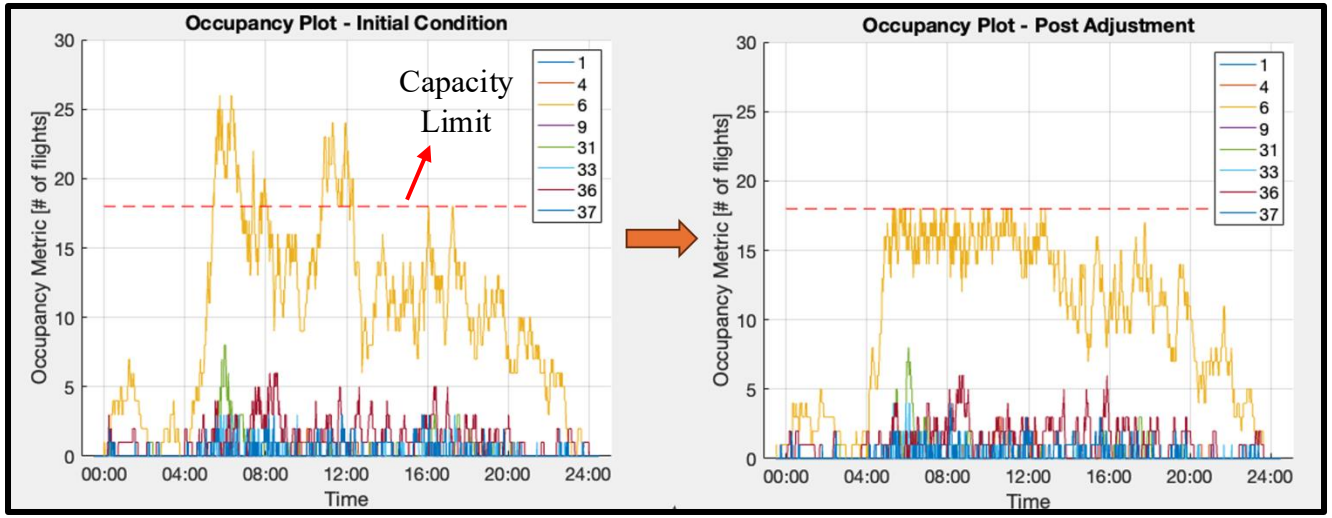
- Centralized coordination struggles with scalability and privacy.
- Agents (aircraft) have conflicting preferences and limited communication.

Proposed Game-theoretic Mechanism



- Each sector adjusts local flight schedules to reduce congestion, guided by a tunable *cooperativeness factor*, κ.
- Best response dynamics ensure *finite-time convergence* to a deterministic strategy (pure Nash equilibrium).

Results



- Experiments with real European flight data show that the proposed algorithm successfully removed the entire overload, whereas baseline methods left residuals.
- Even minimal cooperation ($\kappa = 10^{-6}$) achieves near-centralized performance while maintaining scalability.

Conclusion

Game-theoretic decentralized coordination:

1. enables sectors to mitigate overload through local best responses,
2. guarantees finite-time convergence to stable equilibria,
3. achieves near-centralized performance with minimal cooperation.

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